

# CS341

## ميد الرسم بالحاسب ترم 462

1446/09/13

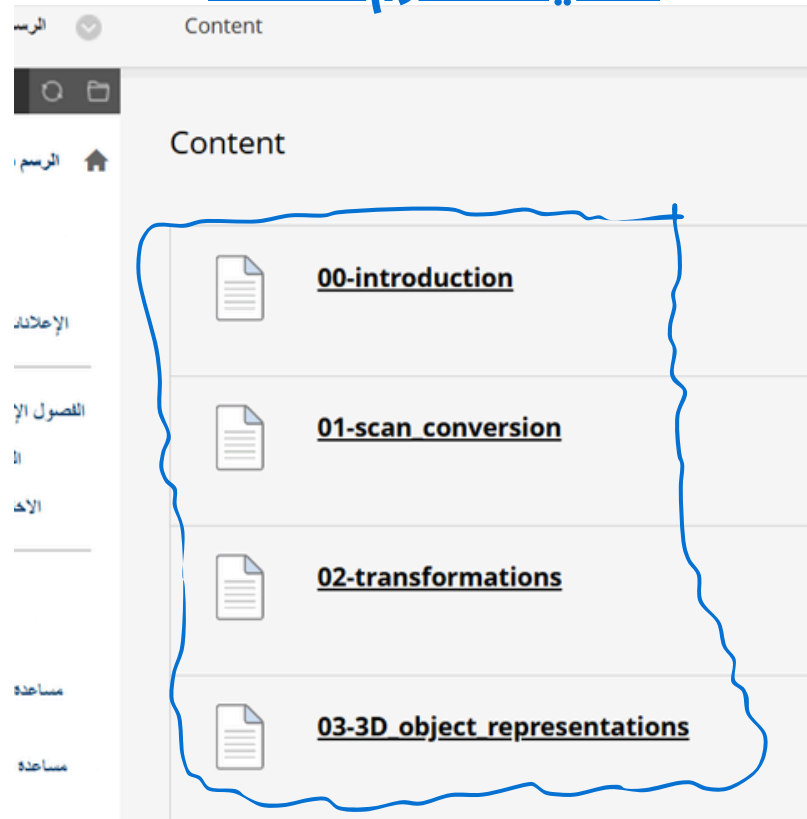
2025/03/13

### الاختبار موحد

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التحديد :

### [سلايدات ترم 462](#)



ادعوا لامي بالشفاء والصحة والعافيه  
وبالتوفيق لكم يارب



# Midterm Exam - Computer Graphics (CS 341)

Second Semester 1446 (462)

Duration: 1 hour

Thursday, 13/3/2025

Student Name: \_\_\_\_\_

Student ID: \_\_\_\_\_

Section #: \_\_\_\_\_

S.N: \_\_\_\_\_

Total Marks: 20

## Question 1: Fill in the gaps.

1. Boundary representations (B-Reps) define the surface of 3D objects using Piecewise Smooth Surfaces like \_\_\_\_\_ or \_\_\_\_\_.
2. Scan conversion is the process of converting \_\_\_\_\_ representations of shapes (e.g., lines, circles) into \_\_\_\_\_ on a raster display.
3. In Bresenham's line algorithm, the decision variable determines whether the direction of the next pixel is \_\_\_\_\_ or \_\_\_\_\_.

## Question 2: Answer the following questions.

A - Mark True or False for each statement

1. Bresenham's line algorithm uses floating-point arithmetic to draw lines. (\_\_\_\_)
2. In raster displays, pixels are arranged in a rectangular grid. (\_\_\_\_)
3. Super-sampling is a method for anti-aliasing in computer graphics. (\_\_\_\_)
4. A voxel model is an example of a boundary representation. (\_\_\_\_)
5. A triangle mesh is preferred in B-Reps. (\_\_\_\_)
6. Computer graphics applications are used only in the entertainment industry. (\_\_\_\_)

## B - Multiple choice questions / Fill in the blanks

1. Which of the following is an example of a geometric transformation that alters the size of the object?

- a) Translation   b) Rotation   c) Scaling   d) Blurring

2. \_\_\_\_\_ is a model heavily used in medical imaging and fluid dynamics.

- a) B-Reps   b) CSG   c) Voxel   d) Triangle Mesh

3. The decision variable in Bresenham's line algorithm determines:

- a) The slope of the line   b) The next pixel to plot   c) The color of the pixel   d) The resolution of the line

4. The implicit equation of a circle is defined as:

- a)  $F(x, y) = x^2 + y^2 - r^2$   
b)  $F(x, y) = mx + n$   
c)  $F(x, y) = ax + by + c$   
d)  $F(x, y) = x + y - r$

5. In transformation matrices, a negative scaling factor is used for:

- a) Translation   b) Rotation   c) Mirroring   d) Shearing

6. What is Constructive Solid Geometry (CSG) used for?

- a) Creating smooth surfaces  
b) Combining simple shapes using Boolean operations  
c) Representing voxel models  
d) Partitioning space into octrees

7. What is Binary Space Partitioning (BSP) used for?

- a) Dividing space into half-spaces using planes  
b) Creating voxel models  
c) Rendering implicit surfaces  
d) Defining polygonal meshes

Question 3: Answer the following questions

A - State two advantages of using triangle meshes in 3D object representation. [2 Marks]

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B - What are the two main types of 3D object representations? Explain briefly. [2 Marks]

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C - Name two applications of scan conversion in computer graphics. [1 Mark]

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Question 4: Answer the following questions

A - Using Concatenated Transformations [2.5 Marks]

Rotate the point  $P=(5,3)$  by 90 degrees around the point  $(1,1)$ .

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B - Based on your knowledge about Bresenham’s algorithm, answer the following: [3 Marks]

Generate the coordinates of the pixels that lie on a line segment having the endpoints  $(3,3)$  and  $(10,7)$  and draw it. (Show the steps of the solution).

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